

Human versus machine as content moderators: examining the effects of moderation strategies on Reddit

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Abstract

This study investigates how cumulative content moderation shapes the longitudinal participation of distinct user groups within online communities. Utilizing a large-scale dataset from Reddit in 2022—comprising 530.5 million engagement records and 13.3 million moderation actions—we examined the interplay between moderation strategy (i.e., hard approaches vs. soft approaches), moderator identity (i.e., human vs. machine), and user heterogeneity (i.e., highly active vs. less active). Results indicate that moderation predominantly influences highly active users; for this group, an increase in the amount of silent removal significantly reduces subsequent engagement. Conversely, soft approaches (e.g., suggestion for users) bolster subsequent engagement, but only when executed by human moderators. Increased automated intervention in a community was associated with a decline in engagement among highly active members. These findings suggest that for community sustainability, the communicative mode and identity of the moderator are more critical than the moderation action itself.

Lay Summary

Online communities rely on rules and moderation to keep discussions safe and constructive. But it is unclear whether frequent moderation helps a group or makes people feel less free and less likely to join in. Using a large dataset from Reddit, we examined how different ways of moderating—and whether a human or a bot does it—change how much people participate. We found that people respond to moderation in different ways depending on how active they are. The highly active users who post and comment the most are the most sensitive to moderation actions. When their posts are silently removed without a reason, these core members post less often afterward. However, when human moderators provide helpful tips and suggestions for future posts, it encourages users to stay active. Moderation done by bots, in contrast, often makes the most active users participate less. These findings suggest that keeping a community lively is not just about enforcing rules. Instead of relying only on bots or silent removals, communities should use human-centered communication to guide and support their members.

The sustainability of online communities depends on active member engagement. However, the openness required for growth also exposes these spaces to disruptive behaviors like incivility and hate speech (Hawkins et al., 2023; Ray et al., 2014). To mitigate these threats and cultivate a positive atmosphere, content moderation has emerged as an essential governance approach, involving the screening of user-generated content to enforce community standards (Schoenebeck & Blackwell, 2021).

While moderation aims to cultivate a sustainable environment, its effectiveness remains empirically debated (Kozyreva et al., 2023). One school of thought suggests that consistent moderation creates a safe, predictable space that encourages

participation by shielding users from harassment (Cosley et al., 2005; Dwork et al., 2024; Wise et al., 2006). This perspective emphasizes moderation as a form of governance that upholds communal norms, promotes trust, and fosters a sense of community. In contrast, critics argue that moderation can be perceived as censorship, creating a “shadow of control” that threatens participant autonomy and leads to disengagement or member churn (Myers West, 2018; Shen et al., 2018).

Existing research has primarily utilized experiments and surveys to uncover how specific moderation actions impact individual attitudes (Matzat & Rooks, 2014; Zhao et al., 2024). However, the external validity of these findings is often limited by the high

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contextual nature of community norms and communicative interactions (Matias, 2019; Shaughnessy et al., 2025). Furthermore, many moderation effects are often cumulative rather than transactional; they develop over time through repeated interactions (Chandrasekharan et al., 2018). To address these gaps, this study shifts from an individual-level perspective to a community-level approach, examining how the frequency of diverse moderation activities affects aggregate user engagement in real-world contexts.

We utilize Reddit as a case study, a platform that relies on a hybrid governance model of human volunteers and automated tools. Specifically, this study investigates the heterogeneous effects of cumulative moderation on community-level engagement across three dimensions: moderation strategy (hard vs. soft), moderator identity (human vs. machine), and user heterogeneity (highly active vs. less active). We introduce the meso-level concept of *community climate* to explain the underlying mechanism across 152 active subreddits.

Literature review and research questions

Community climate, content moderation, and collective engagement

The concept of community climate we propose is derived from organizational climate. Drawing from organizational psychology, it refers to members' collective perceptions of the policies, practices, and procedures that are supported and expected within a given environment (Schneider et al., 2013). This construct provides a robust framework for understanding how shared environments shape collective behavior. The formation of climate perceptions originates at the individual level, primarily driven by a fundamental need to reduce uncertainty. When navigating an ambiguous group context, individuals are motivated to engage in social sensemaking to interpret their surroundings (James & Jones, 1974). Through symbolic social interactions—such as observing how leaders behave and noting which actions are sanctioned—individuals form preliminary climate perceptions (Beus et al., 2023). Crucially, this sensemaking process doesn't require individuals to process every unique stimulus; instead, they employ inference, using salient cues to fill conceptual gaps and construct a coherent perception of the environment (Beus et al., 2023). When individuals within the same environment repeatedly observe consistent symbolic cues—such as the punishments imposed for rule violations—their individual interpretations converge into a shared climate perception (Beus et al., 2023).

Organizational climate is not a monolithic entity but rather crystallizes around specific functional dimensions, most notably autonomy, supportiveness, and procedural justice (Schneider et al., 2013). An autonomous climate reflects the degree of independence and discretion granted to individuals, a supportive climate captures the belief that the organization values member well-being, and a procedural justice climate encompasses the collective judgment that decision-making processes are transparent, unbiased, and fair (Schneider et al., 2013). Armed with these shared perceptions, members develop a collective

understanding of behavioral expectations. This cognitive framework allows members to anticipate the likely consequences of their actions, motivating them to behave in ways that maximize rewards and minimize sanctions (Beus et al., 2023). Consequently, prevailing climate dimensions are intrinsically tied to group-level outcomes: A climate characterized by high autonomy and support is positively related to engagement and commitment, whereas a climate rooted in control and procedural injustice is associated with behavioral withdrawal (Schneider et al., 2013).

We contend that the theoretical mechanisms underpinning organizational climate are highly applicable to informal online communities involving content moderation (McInroy et al., 2021). This applicability rests on the premise that any sustained community, whether a formal organization or a voluntary online group, constitutes a shared social environment. The fundamental driver of climate formation—uncertainty reduction through social sensemaking—is often intensified in online spaces due to inherent features such as lower media richness, asynchronous communication, and reduced social presence (Ferguson et al., 2024; Jiang et al., 2018). New members typically arrive with limited knowledge of a community's implicit norms or behavioral boundaries. To reduce this uncertainty, they actively engage in symbolic interactions: they observe which contributions are sanctioned by moderators, which posts are rewarded with upvotes, and which communication styles are adopted by the majority (Amichai-Hamburger et al., 2016). These observations serve as the salient cues from which individuals draw inferences to construct a perception of “how things are done here.” This emergent climate subsequently guides behavior; for instance, members who perceive a supportive climate are likely to increase their engagement, whereas those who perceive a climate of threat or arbitrariness may withdraw (Jiang et al., 2018; Yang et al., 2017).

Building on this foundation, we formally introduce *community climate* as a theoretical construct to analyze the cumulative effects of online content moderation on community-level collective engagement. We argue that the sustained application of distinct moderation strategies systematically shapes divergent community climates—specifically, the shared perceptual environment regarding autonomy, supportiveness, and procedural justice. These resulting climate perceptions are posited to influence aggregated user behavior, with effects that likely vary based on the intensity of a member's participation.

Content moderation strategies

The definition of content moderation traditionally follows two paths. One stream focuses on the management of harmful content, defining moderation as a reactive practice used by commercial or volunteer agents to intercept incivility, hate speech, or illegality (Oh & Downey, 2025; Weber et al., 2024). A second, more expansive perspective views content moderation as a dynamic practice of social governance and culture-building. In this view, moderation is central to negotiating community norms and the psychological contract between a platform and its members. Accordingly, Grimmelmann (2015) defines content moderation as a governance mechanism that structures engagement to address abuse and promote cooperation.

Similarly, [He et al. \(2025\)](#) and [Ren and Kraut \(2014\)](#) describe it as a regulatory process that manages individual engagement to optimize both the quality and quantity of participation.

Effective content moderation, therefore, extends beyond “hard measures” such as technical removals that enforce compliance to include “soft measures.” These communicative interventions guide members in internalizing community norms, fostering trust, and encouraging prosocial participation ([Gorwa et al., 2020](#)). Following [He et al. \(2025\)](#), we distinguish between two hard measures and three soft measures based on their communicative function.

Hard measures prioritize compliance by restricting deviant behaviors ([Schoenebeck & Blackwell, 2021](#)). On Reddit, this typically involves silent removal (content is deleted without public notification) or public removal (content is deleted accompanied by a public comment). While communities seeking to maintain a “safe space” often rely on these easily implemented tools ([Jiang et al., 2023](#)), their lack of transparency may have unintended consequences.

Soft measures aim to curate discourse through proactive communication, including rules explanation, suggestion for users, and expanded topical conversation ([He et al., 2025](#); [Ruckenstein & Turunen, 2020](#); [Schoenebeck & Blackwell, 2021](#)). A rules explanation provides a retrospective justification for a moderation action, invoking community guidelines to establish legitimacy (e.g., “This post was removed... [because you did not include the location in the title]”). Suggestion for users offers corrective, forward-looking advice to guide future contributions (e.g., “There are instructions in the FAQ as well as our other submission guidelines. Your submission would be more appropriate in r/...”). Finally, expanded topical conversation involves moderators participating as community peers to stimulate positive social interactions ([He et al., 2025](#)). For example, a member asked about an incision infection, and the moderator replied that “it can’t diagnose infection from pictures or symptoms ... cesareans have a 10% infection rate.” By addressing user needs directly, this approach aims to improve dialogue quality and foster sustained participation.

Variability in community climate and user perception

Community climate emerges cumulatively from the sustained application of moderation strategies, with each act of moderation functioning as a recurring signal of the community’s underlying values. Because climate is a shared perception, its formation is likely to be moderated by the characteristics of the user groups observing these signals.

In digital spaces, participation is highly skewed: Approximately 90% of users are lurkers (users who read but do not contribute), while a small fraction are intermittent ($\approx 9\%$) or heavy ($\approx 1\%$) contributors ([Kokkodis et al., 2020](#)). Because the effects of moderation are primarily observable among active participants, this study distinguishes between highly active users (heavy contributors) and less active users (intermittent contributors). Highly active users typically exhibit strong emotional commitment, trust, and a robust sense of community ([Kim et al., 2023](#); [Malinen, 2015](#); [Prom Tep et al., 2022](#)). Conversely, less active users often operate in a transitional stage, testing the culture and

deciding whether to deepen their engagement based on how others are treated ([Amichai-Hamburger et al., 2016](#)).

We argue that the same moderation strategy may be perceived differently across these groups, leading to variable engagement behaviors. Regarding silent removal, a high frequency may negatively influence both groups by undermining the perceived procedural justice climate ([Are, 2025](#); [Schoenebeck et al., 2021](#)). Opaque moderation signals that the community does not value user contributions enough to provide feedback, potentially fostering a climate of unsupportiveness and low autonomy ([Gillespie, 2022](#); [Haimson et al., 2021](#); [Kozyreva et al., 2023](#)). Such perceptions of threat are likely to lead to disengagement and withdrawal among both highly and less active users ([Jiang et al., 2018](#); [Yang et al., 2017](#)).

However, alternative theoretical accounts suggest that silent removal may not always be detrimental. For highly active users, high levels of pre-existing trust and shared norms may buffer them against the adverse effects of opaque moderation ([Malinen, 2015](#)). For less active users, limited attention spans may prevent them from gathering enough signals to form a coherent negative perception of the climate. Furthermore, consistent silent removal might even serve as a signal of order and security, reassuring users that the community is well-maintained and promoting aggregate participation ([Dwork et al., 2024](#)).

Public removal is expected to mitigate the negative climate perceptions associated with silent removal. By providing a public comment, moderators reduce opacity and enhance procedural justice ([Suzor et al., 2019](#)). Similarly, soft measures (explanation, suggestion, and topical conversation) are expected to bolster a supportive and fair climate. These strategies signal that moderation is based on legitimate rules rather than caprice, demonstrating an investment in member growth ([Cai et al., 2024](#); [Ruckenstein & Turunen, 2020](#)). These positive signals may be particularly salient for less active users who are actively scanning the environment for normative cues ([Amichai-Hamburger et al., 2016](#)). But on the other hand, it is also possible that the frequency of these soft interventions may be insufficient to significantly influence the aggregate engagement behavior of less active users. Owing to their limited attention investment in community governance, such users are unlikely to encounter these supportive signals. Additionally, the less active users may function as latecomers who find it difficult to integrate into an already cohesive group ([Amichai-Hamburger et al., 2016](#)). Therefore, even well-intentioned soft measures might fail to influence them.

These different theoretical accounts prompt the following research question:

RQ1: *How do different moderation strategies (silent removal, public removal, rules explanation, suggestion for users, and expanded topical conversation) cumulatively influence the subsequent engagement behavior of highly active users and less active users?*

Variability in community climate shaped by moderator identity

The advancement of machine learning has introduced a new paradigm for community governance: automated moderation at

scale. Platforms like Reddit have increasingly deployed machine-assisted tools to support or replace human volunteer efforts (Jhaver, Birman et al., 2019). These machine moderators are software agents capable of executing management tasks—such as AutoModerator, which removes posts based on pre-defined rules, issues standardized comments, or flags content for human review (Jhaver, Birman et al., 2019; Wright, 2022). As these tools become ubiquitous, it is essential to examine how the frequency of machine versus human interventions differentially shapes the community climate and subsequent member engagement.

Highly active users, characterized by strong emotional commitment and trust, typically expect a supportive and procedurally just environment. However, the delegation of content moderation to machines may undermine these expectations. On one hand, automated systems often operate in a mechanistic, context-insensitive manner (Jhaver, Birman et al., 2019; Song et al., 2023). This technical rigidity increases the risk of “false positives”—erroneous removals that fail to account for linguistic nuance, irony, or community-specific slang. Such errors may be perceived by core members as a violation of procedural justice, signaling that the governance process is indifferent to the complexities of human interaction.

On the other hand, users often harbor inherent psychological biases against automated decision-making. Algorithm aversion suggests that individuals tend to prefer human judgment over algorithmic decisions and lose trust in algorithms more quickly than in humans after observing an error (Burton et al., 2020; Jones-Jang & Park, 2023). This is often compounded by the negative machine heuristic, a mental shortcut where individuals view AI as inherently rigid and incapable of the “logic of care” associated with human judgment (Molina & Sundar, 2024). Consequently, a high frequency of automated intervention may signal a shift toward a control-oriented climate rather than a supportive one, triggering behavioral withdrawal among highly active users (Jiang et al., 2018).

In contrast, human moderators are often perceived as custodians of community culture. Rather than merely enforcing binary rules, they actively navigate the grey areas of social interaction, aiming to stimulate discussion and guide conversations organically (Ruckenstein & Turunen, 2020). The perceived social presence of a human moderator may strengthen the supportive climate, as members interpret human intervention as a sign that the community values their contributions enough to provide personalized attention. However, whether these theoretical differences in moderator identity significantly alter the aggregate engagement of core members remains an open question:

RQ2: *How do highly active users’ future engagement levels vary based on the frequency and type of moderation strategies performed by human versus machine moderators?*

The impact of moderator identity on less active users presents a different theoretical puzzle. On one hand, like their more active counterparts, these users may hold a general preference for human judgment, making them more resistant to machine-led moderation. On the other hand, less active users often operate on the periphery, testing community norms with limited

attention (Amichai-Hamburger et al., 2016). Their relationship with the community is often more transactional and superficial.

This peripheral status may have a dual effect. As learners of community norms, they may be highly sensitive to the tone of moderation but less sensitive to the source. Due to the standardized appearance of many Reddit moderation comments, less active users may not clearly differentiate between a human volunteer and an automated bot. Furthermore, their intermittent engagement may reduce the likelihood that they encounter enough moderation signals to distinguish between moderator identities at all. These combined factors lead to our final research question:

RQ3: *How do less active users’ future engagement levels vary based on the frequency and type of moderation strategies performed by human versus machine moderators?*

Method

Data collection and subreddit selection

This study uses data from Reddit, a prominent social platform with approximately 850 million monthly active users. Reddit’s governance relies on a hybrid of volunteer human moderators and automated tools (see Figure 1). Both human and machine moderators are visible within the community and are distinguished by a “Mod” flair.

We employed a research design with the subreddit-week as the primary unit of analysis. Our sample was derived from the 156 subreddits identified by He et al. (2025) (Table S1). We retrieved all posts and comments within these communities for the 2022 calendar year (January 1 to December 31) via the PushShift API (Baumgartner et al., 2020). The selection of subreddits was guided by three criteria: (a) maturity of machine-assisted moderation (these subreddits have adopted such tools since 2013), (b) sustained community activity, and (c) representativeness across 23 major categories (e.g., Sciences, News & Politics) (Supplementary material A, Table S9).

The 2022 timeframe was selected to avoid potential data quality issues stemming from the June 2023 API policy changes and subsequent platform-wide protests. Additionally, the decade-long presence of automated tools in these subreddits ensures that the effects observed are not confounded with the early-stage innovation diffusion. Four subreddits were excluded due to significant exogenous shocks (e.g., being banned or transitioning to private), resulting in a final sample of 152 subreddits (see Table S2). The raw dataset comprised 503.1 million comments and 27.4 million posts.

Identifying moderation strategies and moderator identities

Moderation actions on Reddit originate from two sources: platform administrators and volunteer subreddit moderators. Admin activities were excluded due to their extreme scarcity ($N=8$; Tables S3 and S4). Furthermore, while moderation can occur via private message, this study focuses on public-facing

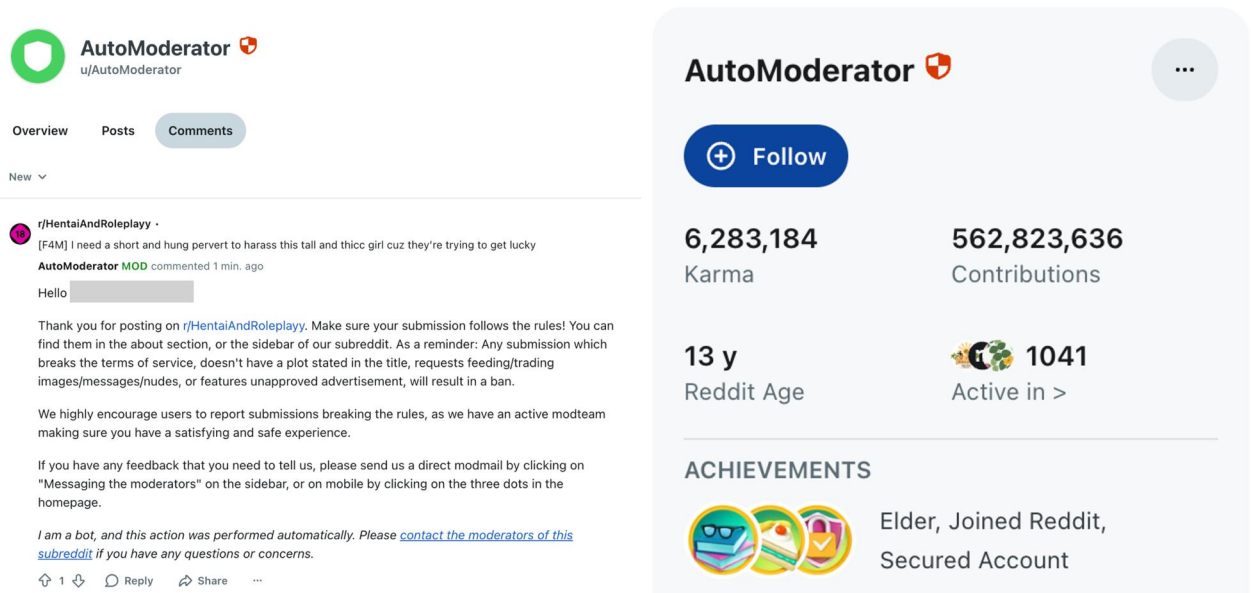


Figure 1 AutoModerator's profile and moderation records. Source: u/AutoModerator, Reddit.

moderation, as private actions do not provide the shared cues necessary for the formation of a collective community climate. Public moderation constitutes over 80% of all regulatory actions on the platform (Jhaver, Appling et al., 2019).

We identified *silent removal* by extracting posts and comments marked as “[removed]” that lacked a corresponding public moderator explanation ($N=9,525,634$). For the remaining four strategies—*public removal*, *rules explanation*, *suggestion for users*, and *expanded topical conversation*—we extracted all content marked with the “moderator” distinction ($N=13,341,838$ for comments, $N=6,375$ for posts).

To classify these strategies, we employed a Bidirectional Encoder Representations from Transformers (BERT) deep learning model. Following the procedures of He et al. (2025), we fine-tuned four binary classifiers. Each moderation message was coded across four non-exclusive dimensions; for example, a single moderator comment could simultaneously indicate a removal, provide a rule-based explanation, and offer a suggestion for future behavior.

To ensure the model's accuracy, we randomly selected 5,000 moderation records from the dataset. Two postgraduate students in communication independently coded 500 records to establish reliability, achieving a Cohen's kappa of 0.95 across the four dimensions. We then used 4,000 labeled records to fine-tune a pre-trained BERT model. Unlike traditional NLP methods, BERT's bidirectional training allows it to capture the contextual meaning of comments regardless of specific word choices (Devlin et al., 2019). Evaluation on a holdout set of 1,000 records demonstrated that both accuracy and recall rates exceeded 85% (see Table S5).

Moderator identities were identified based on usernames or archived activity logs. For instance, machine moderators were identified by names such as “AutoModerator” or explicit self-disclosures (e.g., “I am a bot”). Two coders independently coded the identities of 100 moderators, achieving 100% coding reliability (Cohen's kappa) before coding the remaining data.

Among the 2,103 moderators identified, 325 were classified as bots and 1,152 as human volunteers (see Table S6).

Variables and analysis

Independent variables

Following Hill and Shaw (2021), the present study uses the subreddit-week as its unit of analysis. We aggregated the weekly quantity of five moderation strategies and divided them by the total volume of posts and comments in each subreddit week to calculate moderation rates. The resulting rates were *silent removal* ($M=0.09$, $SD=0.14$), *public removal* ($M=0.07$, $SD=0.63$), *rules explanation* ($M=0.07$, $SD=0.63$), *suggestion for users* ($M=0.07$, $SD=0.57$), and *expanded topical conversation* ($M=0.002$, $SD=0.01$). Notably, these categories are not hierarchically nested; a post or comment may receive an explanation or suggestion without a removal, explaining why the public removal rate may be lower than its associated rhetorical strategies.

We further disaggregated the weekly rates by moderator identity: *public removal* (human: $M=0.004$, $SD=0.02$; machine: $M=0.06$, $SD=0.63$), *rules explanation* (human: $M=0.005$, $SD=0.02$; machine: $M=0.06$, $SD=0.63$), *suggestion for users* (human: $M=0.004$, $SD=0.04$; machine: $M=0.06$, $SD=0.57$), and *expanded topical conversation* (human: $M=0.0004$, $SD=0.005$; machine: $M=0.002$, $SD=0.01$). Due to data limitations regarding the source of silent removal, this variable was treated as a unitary category and was not differentiated by human or machine identity.

Dependent variables

We used two indicators to measure community-level engagement: the weekly count of unique engaged users (those who made at least one post or comment in a given subreddit during that week) and the total number of posts and comments. To examine heterogeneous effects, we differentiated between highly active and less active users. Following Kokkodis et al.

(2020), we classified users based on their post and comment volume throughout 2022 within each subreddit, ranking the top 10% of users in terms of activity as highly active users within that subreddit. By calculating those rankings as such, we maintained a stable cohort of users for each category throughout the observation period. This approach ensures that longitudinal changes in engagement represent shifts in behavior within a consistent group of individuals, rather than “compositional effects” resulting from members moving between activity tiers on a weekly basis. This produced four distinct indicators: the weekly number of engaged highly active users ($M=4,277.78$, $SD=14,571.67$), the number of posts and comments by highly active users ($M=21,221.55$, $SD=87,306.40$), the number of less active users ($M=5,985.33$, $SD=20,195.39$), and the number of posts and comments by less active users ($M=12,472.55$, $SD=44,866.84$).

Control variables

We controlled for two potential confounding factors. First, we calculated each subreddit’s weekly subscriber count by extracting and averaging the values from the “subreddit_subscribers” field for all posts published within that week. Descriptive results indicate that the 152 communities exhibit a wide range of subscriber counts ($M=5,198,394$, $SD=8,485,862$; 25% percentile = 242,102; 75% percentile = 4,168,646). For analysis, we applied a natural logarithm (ln) transformation to these weekly subscriber counts to address skewness. Second, a time variable (week number) was introduced to control for longitudinal trends.

Statistical analysis

Our panel dataset consists of 152 subreddits observed over 53 weeks, totaling 8,056 subreddit-week observations. Each observation in this panel includes 17 variables, which fall into three conceptually distinct groups. First, we constructed five general moderation strategy rates, capturing the overall prevalence of each strategy regardless of moderator identity: silent removal, public removal, rules explanation, suggestion for users, and expanded topical conversation. Second, for four of these strategies (excluding silent removal), we further disaggregated the overall rates by moderator identity, yielding eight moderator-specific strategy rates (four executed by humans and four by machines). These moderator-specific rates are components of the general moderation strategy rates rather than independent constructs. Finally, we included the four dependent variables measuring user engagement.

Before the final analysis, we conducted the necessary diagnostic tests. First, Fisher-type unit root tests on all variables yielded p -values $< .001$, allowing us to reject the null hypothesis that the panels contain unit roots. Second, we conducted a Hausman test to determine the appropriateness of fixed-effects vs. random-effects models. The p -values were $< .001$, which strongly favors the fixed-effects model. This method effectively controls for all time-invariant differences across communities and isolates the effect of within-subreddit variation in moderation activity.

Given our theoretical premise that moderation strategies cumulatively shape the community climate, we examined the effect of each moderation strategy’s weekly volume on aggregated engagement indicators with a one-week lag. We express the basic model as follows:

$$\ln(y_{it} + 1) = \beta X_{i,t-1} + \text{Control}_{i,t-1} + \alpha_i + \varepsilon_{it}$$

where y_{it} is the dependent variable for subreddit i at time t , $X_{i,t-1}$ is a vector of independent variables for subreddit i at time $t-1$; α_i is the unknown intercept (i.e., the fixed-effects); ε_{it} is the random error.

We used clustered standard errors to control the possible autocorrelation within our dependent variables. All models were estimated using the *Stata 16.0 xtreg* package. We tested for multicollinearity among the independent variables; the results indicated a high degree of multicollinearity ($VIF > 5$) when predictors were included simultaneously. To address this issue, we adopted a conservative approach by including each independent variable separately in its own regression model (see [Table S7](#) for detailed model specifications).

Results

Descriptive analysis

The temporal trends of the primary variables were visualized in [Figure 2](#). Overall, the median values across the communities remained relatively stable throughout the study period. However, the mean values exhibited a noticeable decline during the second quarter (Q2). This observation aligns with Reddit’s 2022 Transparency Report, which documented that “removals for content manipulation” in Q2 were approximately 20% below the annual average (see [Table S8](#)). To prevent potential confounding effects from the anomalous decline in moderation activity during week 20, data from that week were excluded from the primary analysis (for detailed justification, see [Supplementary material I](#)).

A comparison of the five moderation strategies revealed that silent removal was the most frequently employed (approximately 10%), while expanded topical conversation was the least utilized. A comparison between human and machine moderators uncovered a substantial disparity in workload ([Figure 3](#)). The frequency of machine-executed removals was approximately 16 times higher than that of human-executed removals. This ratio remained high for explanations and suggestions (approx. 14 times) but lower for expanded topical conversations (approx. four times).

RQ1: The influence of moderation strategies on user engagement

RQ1 asked how distinct moderation strategies influence the engagement of highly active vs. less active users. As shown in Models 1 and 11, a higher silent removal rate was associated with a decrease in both the number of engaged highly active users ($\beta = -0.52$, $SE=0.30$, $p < .1$) and their contribution volume ($\beta = -0.49$, $SE=0.27$, $p < .1$). Similarly, the public removal rate ($\beta = -0.24$, $SE=0.12$, $p < .05$; $\beta = -0.33$, $SE=0.15$, $p < .05$) and rules explanation rate ($\beta = -0.25$, $SE=0.13$, $p < .1$; $\beta = -0.34$, $SE=0.16$, $p < .05$) were negatively associated with highly active users’ engagement metrics. In contrast, neither suggestion for users nor expanded topical conversation significantly impacted highly active users’ engagement.

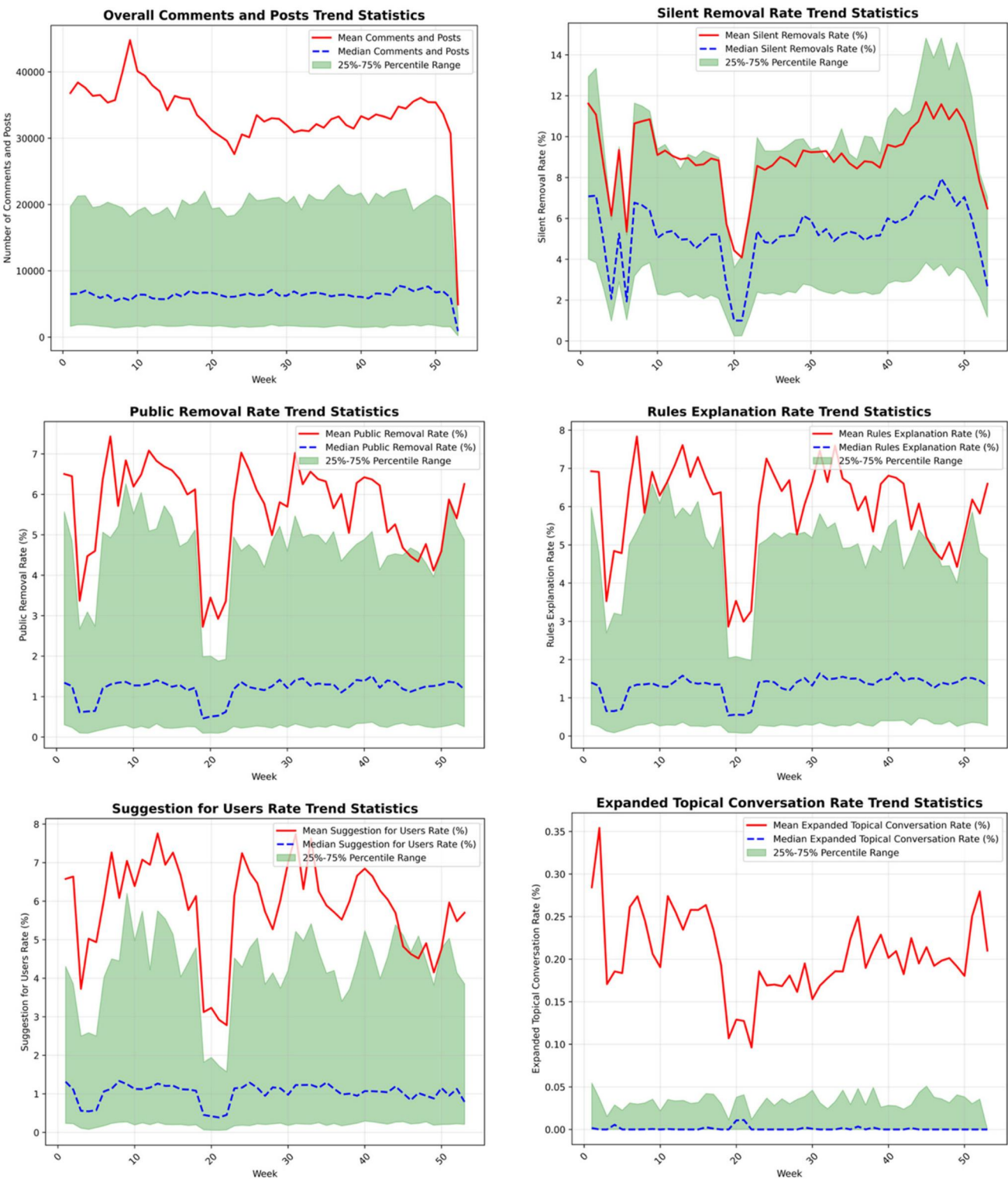


Figure 2 Distributions of five moderation strategy rates and community engagement.

Conversely, as shown in [Tables 1](#) and [2](#), the rates of silent removal, public removal, and suggestion for users showed no statistically significant impact on less active users. Unexpectedly, both a higher rules explanation rate ($\beta = -0.20$, $SE=0.10$,

$p < .05$; $\beta = -0.20$, $SE=0.11$, $p < .1$) and expanded topical conversation rate ($\beta = -1.43$, $SE=0.43$, $p < .01$; $\beta = -1.18$, $SE=0.42$, $p < .01$) were associated with a significant decrease in engagement among less active users.

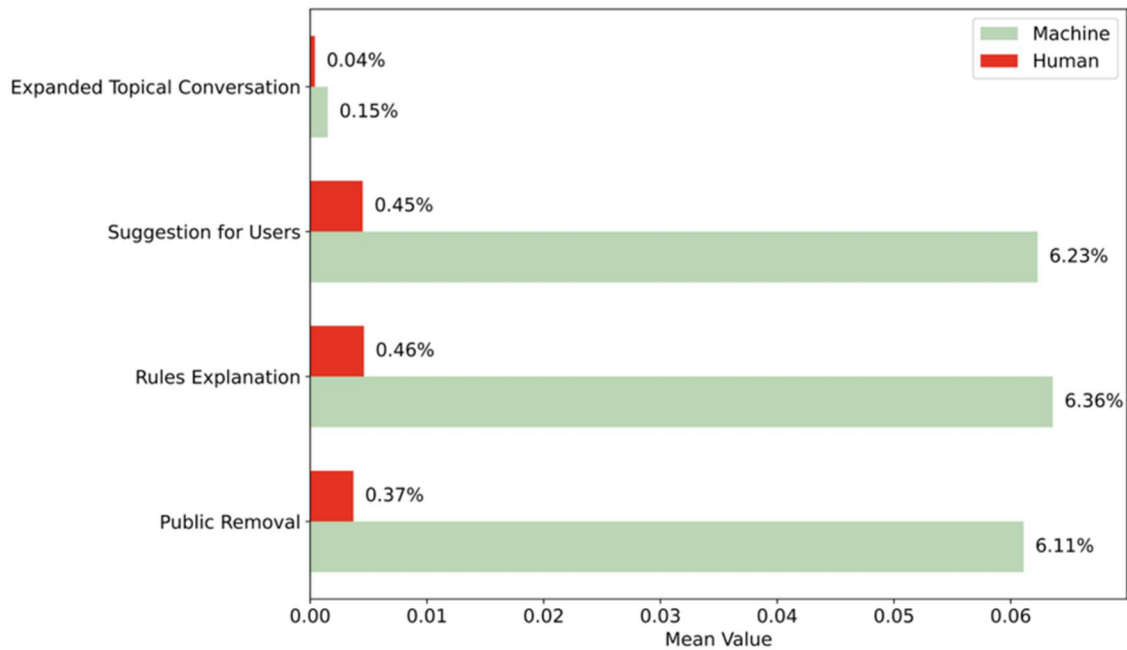


Figure 3 Moderation strategies conducted by machine and human moderator.

Table 1 The impact of moderation strategies on the number of engaged highly/less active users.

Variable	The number of engaged highly active users					The number of engaged less active users				
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
Silent removal rate	−0.52 ⁺ (0.30)					−0.27 (0.22)				
Public removal rate		−0.24 [*] (0.12)					−0.10 (0.13)			
Rules explanation rate			−0.25 ⁺ (0.13)					−0.20 [*] (0.10)		
Suggestion for users rate				−0.15 (0.10)					−0.08 (0.08)	
Expanded topical conversation rate					−0.77 (0.60)					−1.43 ^{**} (0.43)
Constant	−14.69 [*] (5.67)	−14.06 [*] (5.63)	−14.03 [*] (5.64)	−13.99 [*] (5.65)	−13.94 [*] (5.67)	−22.95 ^{***} (5.79)	−22.60 ^{***} (5.82)	−22.63 ^{***} (5.81)	−22.59 ^{***} (5.82)	−22.57 ^{***} (5.83)
Control	Yes									
T	51									
N	152									
R-squared	0.06	0.05	0.05	0.05	0.05	0.08	0.08	0.08	0.08	0.08

*** $p < .001$, ** $p < .01$, * $p < .05$, + $p < .1$. All predictor variables were lagged by one wave ($t - 1$).

RQ2: The impact of moderator identity on highly active users' engagement

Table 3 illustrates the differential impact of moderator identity. Machine-executed strategies including public removal ($\beta = -0.27$, $SE = 0.13$, $p < .05$), rules explanation ($\beta = -0.30$, $SE = 0.16$, $p < .1$), suggestion for users ($\beta = -0.22$, $SE = 0.12$, $p < .1$), and expanded topical conversation ($\beta = -1.62$, $SE = 0.62$, $p < .01$) all exhibited negative effects on the number of engaged highly

active users. Also, these machine-led actions—public removal ($\beta = -0.38$, $SE = 0.16$, $p < .05$), rules explanation ($\beta = -0.41$, $SE = 0.19$, $p < .05$), suggestion for users ($\beta = -0.28$, $SE = 0.14$, $p < .05$), and expanded topical conversation ($\beta = -2.35$, $SE = 0.84$, $p < .01$)—were negatively related to contribution volume.

In contrast, human-executed strategies often foster engagement. A higher rate of human-executed suggestion for users was positively related to the number of engaged highly active users ($\beta = 0.24$, $SE = 0.11$, $p < .05$). Furthermore, human-executed

Table 2 The impact of moderation strategies on the number of posts and comments by highly/less active users.

Variable	The number of posts and comments by highly active users					The number of posts and comments by less active users				
	Model 11	Model 12	Model 13	Model 14	Model 15	Model 16	Model 17	Model 18	Model 19	Model 20
Silent removal rate	−0.49 ⁺ (0.27)					0.05 (0.18)				
Public removal rate		−0.33 [*] (0.15)					−0.15 (0.11)			
Rules explanation rate			−0.34 [*] (0.16)					−0.20 ⁺ (0.11)		
Suggestion for users rate				−0.19 (0.12)					−0.05 (0.06)	
Expanded topical conversation rate					−0.84 (0.98)					−1.18 ^{**} (0.42)
Constant	−22.95 ^{**} (7.03)	−22.43 ^{**} (6.96)	−22.39 ^{**} (6.96)	−22.39 ^{**} (6.98)	−22.32 ^{**} (7.00)	−26.72 ^{***} (6.62)	−26.85 ^{***} (6.68)	−26.84 ^{***} (6.68)	−26.82 ^{***} (6.69)	−26.81 ^{***} (6.69)
Control	Yes									
<i>T</i>	51									
<i>N</i>	152									
<i>R</i> -squared	0.06	0.05	0.06	0.05	0.05	0.09	0.09	0.09	0.09	0.09

*** $p < .001$, ** $p < .01$, * $p < .05$, + $p < .1$. All predictor variables were lagged by one wave ($t - 1$).

Table 3 The impact of moderator identity on highly active users' engagement.

Variable	The number of engaged highly active users				The number of posts and comments by highly active users			
	Model 21	Model 22	Model 23	Model 24	Model 25	Model 26	Model 27	Model 28
Public removal rate by machine	−0.27 [*] (0.13)				−0.38 [*] (0.16)			
Public removal rate by human	0.51 (0.32)				1.14 ^{**} (0.36)			
Rules explanation rate by machine		−0.30 ⁺ (0.16)				−0.41 [*] (0.19)		
Rules explanation rate by human		0.38 (0.26)				0.72 ^{**} (0.26)		
Suggestion for users rate by machine			−0.22 ⁺ (0.12)				−0.28 ⁺ (0.14)	
Suggestion for users rate by human			0.24 [*] (0.11)				0.35 ^{**} (0.12)	
Expanded topical conversation rate by machine				−1.62 ^{**} (0.62)				−2.35 ^{**} (0.84)
Expanded topical conversation rate by human				0.02 (0.27)				0.53 (0.44)
Constant	−13.94 [*] (5.66)	−13.94 [*] (5.65)	−13.94 [*] (5.66)	−13.94 [*] (5.67)	−22.18 ^{**} (7.01)	−22.21 ^{**} (6.99)	−22.31 ^{**} (6.99)	−22.32 ^{**} (6.99)
Control	Yes							
<i>T</i>	51							
<i>N</i>	152							
<i>R</i> -squared	0.05	0.05	0.05	0.05	0.06	0.06	0.05	0.05

** $p < .01$, * $p < .05$, + $p < .1$. All predictor variables were lagged by one wave ($t - 1$).

public removal ($\beta = 1.14$, $SE=0.36$, $p < .01$), rules explanation ($\beta = 0.72$, $SE=0.26$, $p < .01$), and suggestion for users ($\beta = 0.35$, $SE=0.12$, $p < .01$) all exhibited significant positive effects on the contribution volume of highly active users.

RQ3: The impact of moderator identity on less active users' engagement

As shown in Table 4, neither machine nor human-executed public removal and suggestion for users significantly affected the engagement of less active users. However, machine-generated rules explanation showed a marginal negative effect on both the number of engaged less active users ($\beta = -0.20$, $SE=0.11$, $p < .1$) and their contribution volume ($\beta = -0.21$, $SE=0.12$, $p < .1$), while human-executed explanations had no significant effect. Both machine-initiated ($\beta = -1.31$, $SE=0.51$, $p < .05$; $\beta = -1.40$, $SE=0.56$, $p < .01$) and human-initiated ($\beta = -1.54$, $SE=0.70$, $p < .05$; $\beta = -0.97$, $SE=0.57$, $p < .1$) expanded topical conversation negatively affected engagement metrics for this group.

Additional analysis

Robustness checks

We conducted two robustness checks to assess the stability of our results. First, we extended the data aggregation period from one week to two weeks (see Supplementary material D for

details); the key findings remained consistent. Second, to address potential multicollinearity, we re-estimated the models using LASSO and ridge regression. All significant findings were supported by at least one of these penalized regression analyses. To ensure robust inference, we retained the original fixed-effects conclusions that were non-significant in the primary analysis but significant in the penalized models.

Mechanism tests

We posited that the limited responsiveness of less active users may be driven by constrained attention. To test this mechanism, we stratified subreddits into high- and low-frequency groups based on moderation strategy rates (see Supplementary material E for details). Across both user segments, two patterns emerged: (1) strategies significant in the full sample generally remained significant in high-frequency subreddits, but became non-significant in low-frequency subreddits; and (2) strategies non-significant in the full sample typically remained so across both groups. These results suggest that user responses to moderation are substantially attenuated when such actions occur too infrequently to capture collective attention.

Heterogeneous effects

We examined cross-community heterogeneity through three distinct lenses. First, we analyzed heterogeneous effects by community size, as prior research indicates that subreddits differ

Table 4 The impact of moderator identity on less active users' engagement.

Variable	The number of engaged less active users				The number of posts and comments by less active users			
	Model 29	Model 30	Model 31	Model 32	Model 33	Model 34	Model 35	Model 36
Public removal rate by machine	-0.10 (0.13)				-0.15 (0.12)			
Public removal rate by human	-0.27 (0.28)				-0.14 (0.23)			
Rules explanation rate by machine		-0.20 ⁺ (0.11)				-0.21 ⁺ (0.12)		
Rules explanation rate by human		-0.13 (0.18)				-0.01 (0.19)		
Suggestion for users rate by machine			-0.10 (0.10)				-0.07 (0.07)	
Suggestion for users rate by human			0.01 (0.07)				0.03 (0.07)	
Expanded topical conversation rate by machine				-1.31 [*] (0.51)				-1.40 [*] (0.56)
Expanded topical conversation rate by human				-1.54 [*] (0.70)				-0.97 ⁺ (0.57)
Constant	-22.63 ^{***} (5.82)	-22.62 ^{***} (5.81)	-22.57 ^{***} (5.83)	-22.57 ^{***} (5.83)	-26.85 ^{***} (6.69)	-26.81 ^{***} (6.69)	-26.80 ^{***} (6.70)	-26.81 ^{***} (6.69)
Control	Yes							
T	51							
N	152							
R-squared	0.08	0.08	0.08	0.08	0.09	0.09	0.09	0.09

*** $p < .001$, * $p < .05$, + $p < .1$. All predictor variables were lagged by one wave ($t - 1$).

significantly in member diversity, information exposure, moderation practices, and collective identity (Hwang & Foote, 2021). Models were re-estimated for the largest ($M=20,814,722$) and smallest ($M=81,754$) 20% of subreddits in our sample. Results (see [Supplementary material F](#) for details) indicate that in large subreddits, active users are more sensitive to moderation strategies such as silent/public removal, rules explanation, suggestion for users but less responsive to human-led moderation or expanded topical conversation. In small subreddits, the opposite pattern holds: users respond more to human moderation and expanded topical conversation but less to strategies like silent/public removal, rules explanation, and suggestion for users. This is likely due to the stronger social cohesion and higher levels of rule identification characteristic of smaller communities (Hwang & Foote, 2021).

Second, following the methodology of Hill and Shaw (2021), we estimated community-specific models to assess heterogeneity across subreddits, which revealed substantial cross-community variation ([Supplementary material F](#)). For silent removal, public removal, and rules explanation rates, more than half of the subreddits exhibit coefficient directions consistent with the average aggregate estimates. In contrast, for expanded topical conversation, fewer than half of the communities showed effects aligned with the average effect. The relative fragility of the results for expanded topical conversation may be attributed to the very low frequency of this activity in many subreddits, which may limit users' attention and render user responses less consistent. Our mechanism analysis provides partial support for this interpretation.

Third, we examined whether user responses vary by community topic. Overall, consistent with prior research (Lampe et al., 2014), communities focused on News & Politics exhibited the highest sensitivity to moderation among both highly active and less active users. Specifically, strategies that were negatively associated with engagement in the full sample tended to exert stronger negative effects in these communities, while strategies that were non-significant in the aggregate analysis became statistically significant ([Supplementary material F](#)).

Discussion

Key findings

The present study addressed a contentious question in digital governance: How does cumulative moderation impact aggregated user participation within online communities? To explore this effect and delineate its theoretical boundaries, we examined the interplay between moderation strategies, moderator identities, and user groups. Our large-scale analysis yielded several key findings that refine our understanding of community climate and collective behavior (summarized in [Supplementary material C](#)).

First, the silent removal rate negatively impacts engagement specifically for highly active users. For these invested members, whose deep engagement is intrinsically linked to community identity (Kim et al., 2023), opaque moderation may signal a deficit in procedural justice, supportiveness, or autonomy. In contrast, less active users typically maintain a weaker bond with the

community (Amichai-Hamburger et al., 2016). Their participation appears less contingent on shared perceptions of the community climate, rendering them less responsive to normative signals like silent removal. This interpretation is further supported by the absence of effects for public removal and suggestion.

Second, we observed a cumulative negative impact of public removal and rules explanation on highly active users' engagement. While prior research suggests that explanations can boost individual posting frequency (Jhaver Bruckman et al., 2019), our analysis indicates that a substantial portion of this negative aggregate effect is linked to machine execution. When the same strategies are performed by humans, they show no correlation with the number of highly active users but are associated with an increase in posting/commenting volume. This pattern highlights a complex relationship between strategy transparency and agent identity. Although automated moderation formally satisfies transparency requirements, its limited contextual sensitivity and templated nature may render public removal impersonal or punitive. Our manual review revealed instances of machine misclassification ([Supplementary material H](#)), which likely undermine perceptions of support and autonomy. Conversely, human-led public criticism may be perceived as a legitimate social interaction, partially preserving the climate of procedural justice.

Third, the impact of suggestion for users diverges significantly by agent. Machine-executed suggestions were associated with negative effects, whereas human-executed suggestions fostered positive engagement. Moderator identity likely conveys distinct social signals; a machine's suggestion may be perceived as a standardized system directive or a form of external control that threatens the autonomous climate. Conversely, suggestions from human moderators are imbued with greater sociability and interactivity. These are interpreted as personalized, constructive guidance, enhancing the supportive climate and encouraging higher engagement among core members.

Fourth, machine-executed expanded topical conversations were negatively associated with highly active users' engagement, while human-executed versions were non-significant. Manual inspection suggests that machine-initiated discussions are often generic or poorly timed, creating a contextual misfit. For human moderators, the low frequency of this strategy likely reduces its overall saliency, or such actions may simply align with users' existing expectations of human sociability, thus limiting their behavioral impact.

Fifth, machine-executed rules explanations were negatively associated with less active users' engagement. These formulaic, context-poor explanations (Jhaver, Birman et al., 2019) likely fail to convey procedural fairness to newcomers. Instead, they may inadvertently highlight regulatory barriers and entry thresholds, leading less active users to perceive a low-support environment. These experiences may function as deterrence signals, further weakening the already fragile engagement of peripheral users.

Finally, contrary to expectations, expanded topical conversations were negatively associated with less active users' engagement, regardless of the moderator's identity. This suggests a mismatch between high-interaction styles and the needs of low-investment users (Amichai-Hamburger et al., 2016). Human-generated messages in our dataset are often brief or directive, potentially reinforcing power distance and undermining

autonomy. Meanwhile, machine-generated “warm” templates may be perceived as insincere or mechanical, failing to foster an authentic sense of support.

Synthesis of the moderation paradox and community climate

In summary, these findings indicate that the impact of content moderation on user engagement is multifaceted and extends far beyond the traditional concerns of safety and free speech. Our results suggest that engagement is deeply shaped by the broader community climate, specifically through the dimensions of perceived autonomy, supportiveness, and procedural justice. Rather than being a mere technical mechanism for removing objectionable content, visible and communicative moderation functions as a dialogic practice. It conveys community values and provides a framework through which members internalize the platform’s underlying governance philosophy.

Human moderators, in particular, play a key role in enhancing member engagement. By infusing governance with contextual understanding, nuance, and legitimacy, they effectively add warmth to the system and justify the application of rules within specific, complex social situations. These findings highlight a central paradox of the digital age: While automated tools are essential for scaling moderation to millions of users, they remain structurally limited in their ability to foster constructive community climates.

This tension mirrors the fundamental tradeoffs identified in moderation philosophies, where governors must balance nurturing versus punishing approaches, efficiency versus effectiveness, and the quantity of activity versus the quality of contributions (Jiang et al., 2023). Therefore, effective online governance cannot be treated as a one-size-fits-all exercise. It requires navigating tradeoffs by strategically integrating automated efficiency with human judgment and combining hard enforcement with communicative, dialogic strategies to uphold community norms and sustain a healthy, participatory environment.

Theoretical contributions

This study makes several theoretical contributions. First, we introduce a meso-level perspective on content moderation by shifting the unit of analysis from individuals to communities. By examining the cumulative effect of moderation practices on aggregate user engagement, we demonstrate that moderation functions not merely as a tool for addressing individual violations but as a powerful mechanism for shaping the broader community climate.

Second, our study empirically demonstrates that the effectiveness of moderation is contingent upon a tripartite interaction between strategy type, the moderator identity, and user heterogeneity. This helps reconcile previously mixed findings in the literature by showing that the “who” and “how” of moderation matter as much as the “what.”

Third, we extend the concept of organizational climate to the context of large-scale, dynamic online communities. We illustrate its utility in explaining how collective perceptions emerge and influence behavior in computer-mediated spaces,

broadening the scope of organizational research into the digital governance domain.

Limitations and future work

This study has several limitations. First, while we focused on post removal as a primary strategy, other measures, such as private messaging, account suspensions, and temporary mutes, also play significant roles. Future work should examine these alternative moderation strategies if data become accessible. Second, we examined human and machine moderators separately, yet moderation is increasingly a collaborative, hybrid process. Future research should explore user responses to these human-machine partnerships. Third, our model did not control for certain environmental factors, such as toxicity levels, community lifecycles, or interface changes. Fourth, we classified users as highly active or less active based on their total contribution volume throughout the observation period. While this approach ensures a stable cohort, it introduces potential selection and survivorship bias. Because highly active users are defined by their yearlong activity, this group inherently consists of those who were not discouraged enough by the moderation climate to leave the community. Consequently, our findings for this group may reflect the reactions of the most resilient members. Future research should employ rolling-window classification or latent class growth analysis to study how users might transition between activity groups or exit the community entirely in response to specific moderation regimes. Finally, as large language models (LLMs) evolve, the communicative style of machine moderators will become more sophisticated. Future research should examine how LLM-driven moderation reshapes community engagement and behavioral dynamics.

Supplementary material

Supplementary material is available at *Journal of Computer-Mediated Communication* online.

Data availability

The data underlying this article was accessed from the PushShift API. The derived data and code generated in this research are available at <https://osf.io/j3pv4/>.

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Conflicts of interest

None declared.

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